Engine Support Load - Calculation Pilot

Quarto + marimo + Typst feasibility report

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Document control

Field	Value
Document ID	CR-PILOT-001
Revision	0
Status	Pilot - not for design use
Purpose	Test the Quarto marimo engine and Typst output with `calcreport`

1 Basis

This pilot determines the static gravity load from an example engine mass. It tests unit-aware inputs, symbolic calculation display, table output, section references, equation references, and Typst PDF generation.

1.1 Design inputs

$$m_{\text{engine}} = 3448\,\mathtt{kg}$$

Engine mass

$$g = 9.81\,\mathtt{m}/\mathtt{s}^2$$

Acceleration due to gravity

2 Calculation

Using the inputs defined in Section 1, the engine weight is calculated below.

$$F_{\text{engine}} = m_{\text{engine}} \cdot g = 33.82\,\mathtt{kN} \tag{1}$$

Static engine weight

The calculated load is nominally referenced as Equation 1.

3 Results

Table 1: Calculation summary

Item	Value
Engine mass	3448 kg
Gravity	9.81 m/s^2
Engine weight	33.82 kN

The results are summarised in Table 1.

4 Conclusion

The example static engine load is **33.82 kN**. This is a workflow pilot only; it is not an issued engineering calculation.